

Innovation under Uncertainty: Bolstering R&D, Patenting via Secondary Markets

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The following is based on joint work with Andrew Toole, Gerard Torres, and Richard Miller, all of the *U.S. Patent and Trademark Office (USPTO)*, *Office of the Chief Economist (OCE)*.

Introduction

Economic Upheavals

- Coronavirus Pandemic 2020 -
- Great Recession 2008-9
- Internet Bubble 2000-1

Innovation resilience

Three takeaways during COVID-19 Experience:

- Declines in output in IP-industries *fared better* than in non-IP-industries
- Investments in knowledge (R&D) more *stable* than investments in physical capital
- IP protection allows for *reversible* investment – that is, companies can store value and recoup investments in knowledge (but this option varies by technological field)

Outline

Goals:

- Explain resiliency of innovation under uncertainty (potential for reversible investment)
- Measure (i) economic uncertainty in different ways and (ii) exposure to secondary IP markets

Plan of Talk:

- Discuss some trends (related to pandemic shock)
- Review real options concepts
- Present evidence more generally (i.e., for shocks in general)
 - Empirical framework
 - Measures of uncertainty and Index of Secondary Patent Market exposure
 - Data sources
 - Test results
- Discuss implications and way forward

Key findings:

- Confirm the key takeaways: innovation is 'special' and resilient under shocks.
- Patenting is normally *negatively* associated with uncertainty.
- But for firms with sufficient access to secondary patent markets, increased uncertainty could spur greater patenting

Related Literature

Uncertainty and investment

- Abel et al. (1996) "Options, the Value of Capital, and Investment." QJE
- Bloom et al. (2022) "Investment and Subjective Uncertainty." NBER
- Bulan (2005) "Real Options, Irreversible Investment and Firm Uncertainty." RFE
- Dixit and Pindyck (1994) *Investment Under Uncertainty*. PUP

Covid effects on business, employment, and innovation

- Buffington et al. (2021) "High Frequency Business Dynamics ..." Census Bureau
- Davis et al. (2021) "STEM Employment Resiliency ..." NBER
- Dinlersoz et al. (2021), "Business Formation ..." Census Bureau

Business shocks and patenting

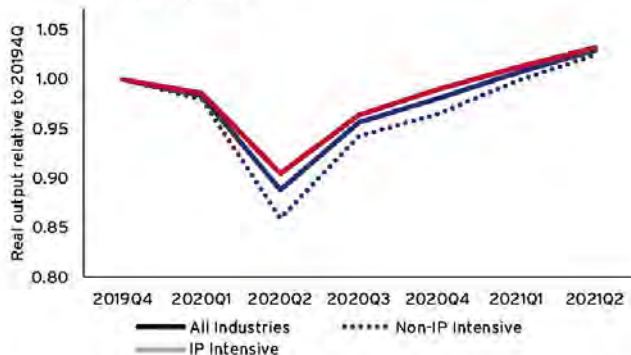
- Bloom and Van Reenan (2000) "Patents, Real Options, and Firm Performance." EJ
- Czarnitzki and Toole (2011) "Patent Protection, Market Uncertainty, and R&D ..." ReSTAT
- Hingley and Park (2017) "Do Business Cycles Affect Patenting? ..." TFSC

Secondary markets for patents

- Akcigit et al. (2016) "Buy, keep, or sell ... market for ideas." ECTA
- Love et al. (2018) "An Empirical Look at the 'Brokered' Market ..." MLR
- Oliver et al. (2019) "Tracking Buyer and Seller Behavior ..." IPWatch
- Serrano (2018) "Estimating the Gains from Trade in the Market for Patent Rights." IER.

Covid-19 Shock: U.S.

FIGURE 1 US REAL GROSS OUTPUT



Note: Gross real output by Industry. The gross output measure, seasonally adjusted at annual rates, is used, and converted into real 2012 dollars using its chain-type price index.

Source: Authors' calculations based on US Bureau of Economic Analysis Interactive Tables (www.bea.gov).

Tangible vs. Intangible Investments

TABLE 1 INVESTMENT RATES: PHYSICAL CAPITAL INVESTMENT VERSUS R&D

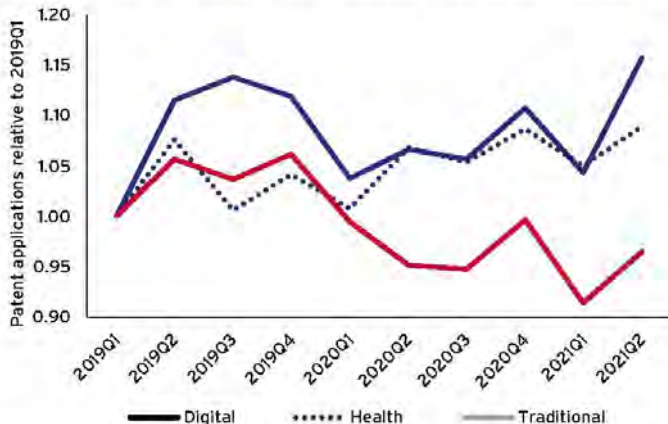
	Rate of physical capital investment to sales	Rate of R&D investment to sales
2019:Q1	10.3	7.7
2019:Q2	1.9	7.4
2019:Q3	-1.6	7.3
2019:Q4	14.9	7.5
2020:Q1	27.6	8.1
2020:Q2	-12.3	8.5
2020:Q3	-4.5	7.8
2020:Q4	17.2	7.6
2021:Q1	-3.4	7.8
2021:Q2	3.3	7.6
2021:Q3	1.8	7.5
<i>Mean</i>	5.0	7.7
<i>Standard deviation</i>	11.5	0.3
<i>Coefficient of variation</i>	2.3	0.04

Notes: R&D is research and development expense; physical investment is the change in the stock of capital. The mean, standard deviation, and coefficient of variation are calculated over the period shown. The coefficient of variation equals the ratio the standard deviation to the mean, an indicator of the volatility (or spread) in the investment rates relative to the average investment rate. The bolded rows mark the period of the initial COVID-19 shock on business operations.

Source: S&P Capital IQ (www.capitaliq.com).

Industry Differentiation

FIGURE 2 PATENT APPLICATIONS FROM US-BASED INVENTORS BY INDUSTRY GROUP



Notes: Utility (Invention) patent applications filed by receipt date, the date which the application is initially submitted to USPTO. The first-listed CPC symbol for each application is used to classify each application. Origin counts are fractionalised based on each inventor's country code in their mailing address.

Source: Author's calculations based on data from USPTO's Patent Location and Monitoring system (PALM).

Secondary Market Deals

TABLE 2 SECONDARY MARKET DEALS IN PATENT RIGHTS: INDUSTRY REPRESENTATION

Industry group	Counts	Share (%)
Digital technologies	393	44.0
Health technologies	116	13.0
Automotive and parts	65	7.3
Telecommunications	46	5.2
Manufacturing	46	5.2
Media & entertainment	31	3.5
Electricity	27	3.0
Transport	27	3.0
Security & defence	26	2.9
Household goods	22	2.5
Chemicals	13	1.5
Food, tobacco, & beverages	13	1.5

Oil & gas	10	1.1
Finance	10	1.1
Retail & distribution	10	1.1
Construction & building materials	7	0.8
Test and measurement	7	0.8
Support services	6	0.7
Steel & other metals	5	0.6
Utilities & mining	5	0.6
Leisure & hospitality	4	0.4
Printing	3	0.3
Education	1	0.1
TOTAL	893	100.0

Note: Counts of transactions for all technology categories. Each transaction, or deal, consists of one or more patents.
Source: IAM Market (<https://portal.iam-market.com/browse-ip-tech>, accessed 21 March 2022).

Possible Reasons for Trends

- Shocks and lockdowns differentially hurt companies relying on **f2f business** relationships.
- Investments in knowledge are directed to **longer-term** objectives.
- Smoothness of R&D due partly to **adjustment costs** (employment of high-skilled labor).
- Investments in knowledge can be repurposed. They're **not** necessarily **sunk** costs.
- IP can be used to store value during uncertain times. IP is **tradable** in the future.
- Heterogeneity exists across **technological fields**, such as digital technologies, technology hardware and software.

Real Options Theory: Lessons

Developed for Real Investments ... but has implications for Intangible Investments

- Investment is an intertemporal decision
- Future capital investment can be expanded, contracted, or be irreversible
- Future returns may be uncertain
 - When firms can resell capital at a later date, it possesses a *put option*
 - When firms can purchase capital at a later date, it possesses a *call option*
 - Both options affect current incentives to invest
 - Higher values of the call option create incentives to postpone investment, while higher values of the put option create incentives to invest currently
- Greater future uncertainty can have an ambiguous effect on the incentive to invest because it increases the value of both the put and call options.
- Patent resale markets allow innovation investments to be reversible. The more reversible, the greater the incentive to innovate and patent.
- Diminished opportunities for resale and higher uncertainty may delay patenting or lead to patent abandonment.

Estimating Equation

Basic model of patentable innovation:

$$P_{it} = \exp(\alpha_0 + \alpha_i + \alpha_t + \beta_1 UNC_{it} + \beta_2(ISE_i \times UNC_{it}) + \mathbf{X}'_{it}\delta)\varepsilon_{it},$$

- Unit of analysis is company i at quarter t
- P_{it} denotes patent grants by date of application
- UNC_{it} is the level of uncertainty faced by firm i at time t
- ISE_i is the long term exposure of firm to the secondary patent market.
- $\mathbf{X}_{it}$ is the vector of time-varying firm controls: R&D, sales, ratio of cash flow to capital stock, assets, firm age, industrial concentration (HHI of sales share in industry), and technology specialization (HHI of CPC code share in firm's patent portfolio).
Not all reported in tables below to conserve space
- α_i and α_t are firm and quarter fixed effects.
- ε_{it} is the stochastic error term.

Exponential Model ($\exp$) is estimated by PPML (w/ high-density fixed effects). Method of estimation is robust to any heteroskedasticity in ε_{it} and to zero counts of P_{it} .

Hypothesis: $\beta_1 < 0$ and $\beta_2 > 0$

Alternative Approaches to Measuring Uncertainty and Shocks

(1) Coefficient of variation:

$$UNC_{it} = \frac{\sqrt{\text{var}(z_{it})}}{\text{mean}(z_{it})}$$

$$\text{where } z_{it} = \frac{\frac{\text{sales}_{it}}{\text{employees}_{it}}}{\frac{1}{N^{\iota}} \sum_{i=1}^{N^{\iota}} \frac{\text{sales}_{it}}{\text{employees}_{it}}}$$

and N^{ι} = number of firms in industry ι .

(2) Auto-regressive model:

$$z_{it} = \gamma_0 + \gamma_1 z_{it-1} + \gamma_2 z_{it-2} + \gamma_3 z_{it-3} + \gamma_4 z_{it-4} + \gamma_t t + \epsilon_{it}$$

From the standard deviation of the residuals: $UNC_{it}^{AR} = \sigma(\epsilon_{it})$.

Alternative Approaches to Measuring Uncertainty and Shocks

(3) Stock Market Volatility:

Let S denote stock price and $\rho = \frac{\Delta S + \text{DIV}}{S}$ the rate of return. Regress firm rate of return on those of the market and industry (where $\rho_{I\tau}$ is made orthogonal to $\rho_{M\tau}$):

$$\rho_{i\tau} = \beta_{it} + \beta_{it}^M \rho_{M\tau} + \beta_{it}^I \rho_{I\tau} + \epsilon_{i\tau}$$

From the standard deviation of the residuals: $UNC_{it}^{\text{Stk Volat.}} = \sqrt{n} \epsilon(\rho_{it})$, where n = number of trading days, τ , per quarter.

(4) Cyclical-Trend Filtering (Baxter-King method):

Let $z_{it} = z_{it}^T + z_{it}^C$, where z_{it}^T denotes the trend level of real sales per worker and the z_{it}^C the cyclical component.

Business shock variable, $BK_{it} = \frac{z_{it}^C}{z_{it}^T}$

Brokered Markets

Using data on patent 'packages', we obtain **prices** at which packages are sold and determine **technological fields (CPC codes)** covered by the patents.

We exploit idea that (a) some technological fields are more involved than others in trading and (b) firms vary in the technological fields specified in their patent applications.

The *Index of Secondary Patent Market Sales Exposure*, *ISE* for firm i is:

$$ISE_i = \sum_{j=1}^N Share_{ij} \times Proportion_j$$

- j indexes technological field and N the total number of technological fields the firm innovates in during a given sample period.
- $Share_{ij}$ is the share of the j^{th} technological field in the firm's overall patent grants.
- $Proportion_j$ is the share of the j^{th} technological field in all the patent packages.

Two versions:

- 1 $Proportion_j$ is *unweighted*, based on raw counts: $\frac{CPC_j}{\sum_j CPC_j}$
- 2 $Proportion_j$ is *weighted*, by value of an imputed price for field j : $\frac{p_j CPC_j}{\sum_j p_j CPC_j}$

Table 3: Top 12 CPC subclasses in the share of secondary market sales of granted patents

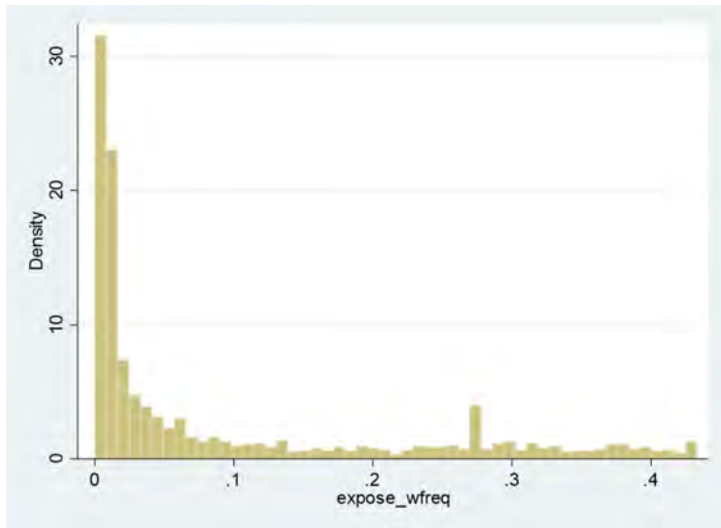
CPC Sub Class	% share total value	CPC_SubClass_Title
H04	43.29	Electric Communication Technique
G06	27.20	Computing; Calculating; Counting
H01	6.62	Basic Electric Elements
G01	2.98	Measuring; Testing
H03	2.67	Basic Electronic Circuitry
G11	2.07	Information Storage
G02	1.69	Optics
G08	1.67	Signaling
A61	1.52	Medical or Veterinary Science; Hygiene
G10	1.50	Musical Instruments; Acoustics
H05	1.01	Electric Techniques not Otherwise Provided for
H02	0.90	Generation; Conversion, Distribution of Electric Power

Table 4: Sample Indexes of Secondary Market Sales, by Industry

NAICS code	Industry	Index of Sales Exposure
5182	Telecomm Data Process. Services	0.3007
5112	Software publishers	0.2602
3342	Communications equipment manuf.	0.2062
3343	Audio and video equipment manuf.	0.1962
3341	Computer and peripheral equipment manuf.	0.1959
3344	Semiconductors and other elect. comp. manuf.	0.1209
5324	Commercial and Industrial Machinery...	0.0720
2211	Electric Power Gen., Trans, & Distr.	0.0481
3364	Aerospace products and parts manuf.	0.0347
3332	Industrial machinery	0.0340
3272	Glass and glass products	0.0338
3391	Medical equipment and supplies manuf.	0.0264
3361	Motor vehicles manufacturing	0.0261
3336	Engines, turbines, and power trans. equip. manuf.	0.0199
3254	Pharmaceuticals and Medicines	0.0153
3262	Rubber Products	0.0108
3261	Plastic Products	0.0070
3311	Iron and Steel Mills and Ferro alloys	0.0067
3251	Basic Chemicals	0.0059
6216	Home Health Care Services	0.0010

Index of Secondary Market Exposure (ISE)

Figure 3: Histogram



Sample: Companies filing at USPTO

Firm Level Data: 1990 - 2021, quarterly

① Major Patenting Firms in the United States:

- 800 companies (*accounting for 40% of all U.S. grants*)
- 15 countries represented, including U.S.
- 94 industries, NAICS 4-digit level

② Data Sources:

- AIR and USPTO *PatentsView*
- Standard & Poor *Capital IQ*
- Walls & Assoc. *National Establishment Times-Series (NET)* database
- IAM market, ROI Insights
- U.S. Bureau of Economics Analysis *Data by Economic Accounts*.

③ Data Preparation:

- Firm name disambiguation
- Concordance: match assignees/subsidiaries to parent companies
- Match patenting data to firm level data in *Capital IQ*
- Assignment records not available in pre-grant data. Hence, we use data on patent grants as of the date of filing.
- All relevant data converted to real 2012 dollars

Table 5: Main Results

	Grants by Filing Date					
	(1)	(2)	(3)	(4)	(5)	(6)
Ln R&D Flow	0.213*** (0.027)	0.214*** (0.027)	0.215*** (0.027)	0.214*** (0.027)	0.219*** (0.027)	0.218*** (0.027)
Ln Sales	0.107** (0.042)	0.118*** (0.042)	0.114*** (0.042)	0.105** (0.042)	0.111*** (0.042)	0.112*** (0.042)
Indus Concentration	-0.197 (0.141)	-0.282** (0.140)	-0.245* (0.141)	-0.187 (0.140)	-0.288** (0.140)	-0.270* (0.140)
Tech Specialization	-0.765*** (0.060)	-0.784*** (0.060)	-0.774*** (0.060)	-0.763*** (0.060)	-0.788*** (0.060)	-0.781*** (0.060)
UNC (Sales)	-0.157*** (0.054)	-0.501*** (0.110)	-0.428*** (0.122)			
UNC (Sales) $\times$ ISE		2.478*** (0.501)				
UNC (Sales) $\times$ Unweighted ISE			1.950*** (0.591)			
UNC (Profit)				-0.147*** (0.049)	-0.405*** (0.095)	-0.411*** (0.101)
UNC (Prof) $\times$ ISE					1.739*** (0.376)	
UNC (Prof) $\times$ Unweighted ISE						1.784*** (0.422)
Constant	2.130*** (0.317)	2.170*** (0.305)	2.149*** (0.310)	2.118*** (0.317)	2.118*** (0.307)	2.112*** (0.309)
Threshold Index		0.20	0.22		0.23	0.23
Pseudo R^2	0.90	0.90	0.90	0.90	0.90	0.90
Observations	23,161	23,161	23,161	23,156	23,156	23,156

Table 6: Alternative Measures of Uncertainty

	Grants by Filing Date					
	(1)	(2)	(3)	(4)	(5)	(6)
Ln R&D Flow	0.285*** (0.027)	0.279*** (0.027)	0.201*** (0.025)	0.200*** (0.025)	0.281*** (0.026)	0.283*** (0.026)
Ln Sales	0.101** (0.043)	0.094** (0.043)	0.138*** (0.040)	0.133*** (0.040)	0.119*** (0.042)	0.115*** (0.042)
UNC ^{AR}	-0.111*** (0.029)	-0.345*** (0.077)				
UNC ^{AR} × ISE		1.170*** (0.308)				
UNC_Stock Volat.			-0.006** (0.003)	-0.016*** (0.004)		
UNC_Stock Volat. × ISE				0.083*** (0.025)		
Cycl. shock (BK)					-0.101*** (0.032)	-0.257*** (0.043)
Cycl. shock (BK) × ISE						1.380*** (0.298)
Constant	2.646*** (0.312)	2.623*** (0.311)	1.948*** (0.272)	1.908*** (0.272)	2.000*** (0.326)	2.081*** (0.305)
Threshold Index		0.29		0.19		0.19
Pseudo R ²	0.90	0.90	0.89	0.90	0.91	0.91
Observations	22,068	22,068	23,847	23,847	21,684	21,684

Table 7: By Select Sectors

	Grants by Filing Date			
	INFO TECH	INFO TECH	BIO- PHARM	BIO- PHARM
	(1)	(2)	(3)	(4)
Ln R&D Flow	0.280*** (0.039)	0.276*** (0.040)	0.363*** (0.063)	0.368*** (0.063)
Ln Sales	0.069 (0.054)	0.084 (0.054)	0.015 (0.093)	0.019 (0.093)
UNC	-0.192*** (0.056)	-0.914*** (0.119)	-0.622*** (0.147)	-0.739*** (0.188)
UNC $\times$ ISE		4.056*** (0.563)		8.761 (5.639)
Other CONTROLS	Y	Y	Y	Y
Constant	1.040*** (0.362)	1.124*** (0.349)	4.223*** (0.582)	4.187*** (0.577)
Threshold Index		0.22		n/a
Pseudo R^2	0.90	0.90	0.77	0.77
Observations	8,999	8,999	2,962	2,962

Table 8: By Firm Size

	Grants by Filing Date			
	Below Median	Below Median	Above Median	Above Median
	(1)	(2)	(3)	(4)
Ln R&D Flow	0.020 (0.040)	0.026 (0.039)	0.293*** (0.030)	0.291*** (0.029)
Ln Sales	0.138** (0.055)	0.170*** (0.055)	0.123** (0.050)	0.130*** (0.050)
UNC	-0.452*** (0.135)	-1.197*** (0.251)	-0.110** (0.052)	-0.409*** (0.105)
UNC $\times$ ISE		4.161*** (0.990)		2.206*** (0.514)
Other CONTROLS	Y	Y	Y	Y
Constant	-0.259 (0.615)	-0.154 (0.595)	2.188*** (0.337)	2.223*** (0.327)
Threshold Index		0.29		0.19
Pseudo R^2	0.80	0.80	0.89	0.89
Observations	11,615	11,615	11,546	11,546

Table 9: Quasi-simulation: Period 2017 - 2021

Shock to Revenue = 10% of Std Deviation

Companies	Revenue Change %	Uncertain. Change %	Exposure Index	Total New Grants	Indirect New Grants	Original Grants
Bristol-Myers	3.40	-3.32	0.009	644	506	279
Gilead	4.43	1.57	0.010	126	72	108
Microsoft	3.52	0.56	0.299	5837	4031	3747
Qualcomm	4.31	-0.74	0.314	4945	3148	3729

Indirect effect measures revenue change on patenting via uncertainty

Total effect includes direct effect of revenue change on patenting

Recap

- Results consistently show $\beta_1 < 0$ and $\beta_2 > 0$. That is, uncertainty has a **negative** association with patenting, but above a threshold level of secondary patent market exposure, uncertainty has a **positive** association with it. Threshold varies by firm size and sector.
- A priori, uncertainty has diverse potential effects, which can be associated with higher revenues or lower revenues (namely, the **volatility**).
- Positive implication: study explains **resilience** or relative stability of innovation over the business cycle or under fluctuating uncertainty.
- (Potential) Normative implication: expand **secondary markets for patents**. Improve their efficiency. Formalize like equity markets.
- Future research could investigate **other factors** that make innovation resilient: STEM education and training, institutional support (IP offices), labor market for R&D personnel, organizational and managerial competence, public policy support, and networks (regional and international).
- Future work could also try to **replicate** the study for other countries; e.g., Indian firm-level data (if secondary patent markets exist and data on sales are available)